# Practice 8 — Recurrence Relations

Solutions to Selected Problems

Problems 2, 6

Discrete Mathematics — QYSU

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# 1 Problem 2 — Solving a Second-Order Linear Recurrence

**Theory Recap: Second-Order Linear Recurrences with Constant Coefficients**

A **second-order linear recurrence with constant coefficients** has the form

$$a_n = c_1\, a_{n-1} + c_2\, a_{n-2}, \qquad n \geq 2,$$

where $c_1, c_2$ are constants and $a_0, a_1$ (or $a_1, a_2$) are given initial values.

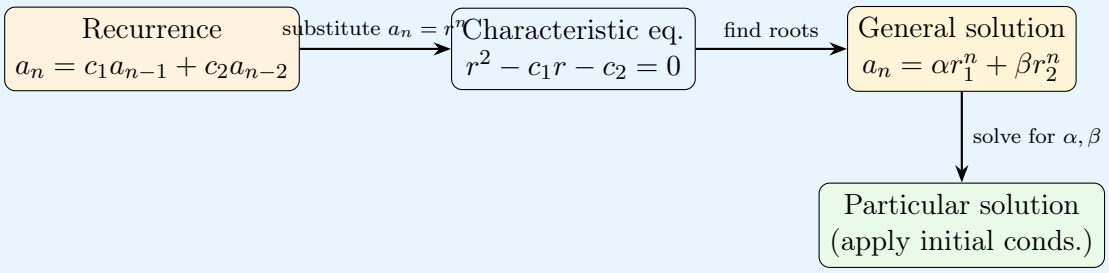
**The Characteristic Equation Method.** Substitute the trial solution $a_n = r^n$ into the recurrence to obtain the **characteristic equation**:

$$r^2 - c_1\, r - c_2 = 0.$$

The form of the general solution depends on the roots $r_1, r_2$:

| Case | Roots | General Solution |
|---|---|---|
| Distinct real roots | $r_1 \neq r_2$, both real | $a_n = \alpha\, r_1^n + \beta\, r_2^n$ |
| Repeated real root | $r_1 = r_2 = r$ | $a_n = (\alpha + \beta\, n)\, r^n$ |
| Complex conjugate roots | $r_{1,2} = \rho\, e^{\pm i\theta}$ | $a_n = \rho^n \big(\alpha \cos n\theta + \beta \sin n\theta\big)$ |

The constants $\alpha$ and $\beta$ are determined by substituting the initial conditions into the general solution and solving the resulting $2 \times 2$ linear system.




**Problem Statement**

Solve the linear recurrence relation

$$a_n = 5a_{n-1} - 6a_{n-2}, \qquad n > 2,$$

with initial conditions $a_1 = 2$ and $a_2 = 10$.

## Solution

**Step 1: Write the characteristic equation.**

The recurrence $a_n - 5a_{n-1} + 6a_{n-2} = 0$ has characteristic equation

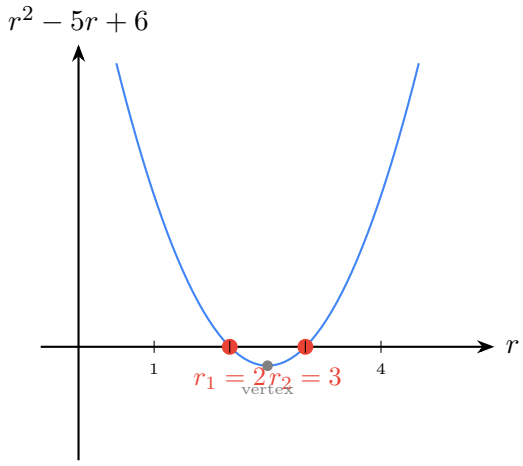
$$r^2 - 5r + 6 = 0.$$

**Step 2: Find the roots.**

Factoring:

$$r^2 - 5r + 6 = (r-2)(r-3) = 0 \qquad \Longrightarrow \qquad r_1 = 2, \quad r_2 = 3.$$

The roots are distinct and real, so we are in **Case 1**.




**Step 3: Write the general solution.**

Since $r_1 = 2$ and $r_2 = 3$ are distinct real roots:

$$a_n = \alpha \cdot 2^n + \beta \cdot 3^n.$$

**Step 4: Apply initial conditions.**

Substituting $n = 1$ and $n = 2$:

$$\begin{cases} a_1 = 2\alpha + 3\beta = 2 \\ a_2 = 4\alpha + 9\beta = 10 \end{cases}$$

From the first equation:

$$\alpha = \frac{2 - 3\beta}{2} = 1 - \frac{3\beta}{2}.$$

Substituting into the second equation:

$$4\left(1 - \frac{3\beta}{2}\right) + 9\beta = 10 \quad \Longrightarrow \quad 4 - 6\beta + 9\beta = 10 \quad \Longrightarrow \quad 3\beta = 6 \quad \Longrightarrow \quad \beta = 2.$$

Then $\alpha = 1 - \dfrac{3 \cdot 2}{2} = 1 - 3 = -2$.

**Step 5: Write the particular solution.**

$$a_n = -2 \cdot 2^n + 2 \cdot 3^n = 2\big(3^n - 2^n\big).$$

**Step 6: Verify.**

| $n$ | $2(3^n - 2^n)$ | Expected | Match |
|---|---|---|---|
| 1 | $2(3-2) = 2$ | $a_1 = 2$ | ✓ |
| 2 | $2(9-4) = 10$ | $a_2 = 10$ | ✓ |
| 3 | $2(27-8) = 38$ | $5(10) - 6(2) = 38$ | ✓ |
| 4 | $2(81-16) = 130$ | $5(38) - 6(10) = 130$ | ✓ |

$$\boxed{a_n = 2\left(3^n - 2^n\right)}$$

## 2 Problem 6 — Binary Strings with Two Consecutive Zeros

**Theory Recap: Complementary Counting and Recurrences for Constrained Strings**

**Complementary counting.** When counting objects that satisfy a property $P$, it is sometimes easier to count those that do *not* satisfy $P$:

$$|P| = |\text{total}| - |\text{not } P|.$$

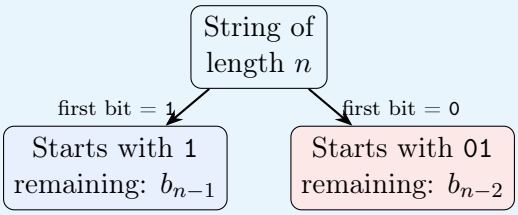
**Setting up recurrences for constrained strings.** To count binary strings of length $n$ satisfying (or avoiding) a pattern:

1. Define a sequence: let $b_n$ be the count of valid strings of length $n$.
2. **Condition on the first (or last) character.** Each choice constrains what can follow, yielding a recurrence.
3. Solve using base cases.

**Example pattern: avoiding "00".** For a string of length $n$ with no two consecutive 0s:

- If it starts with `1`: the remaining $n-1$ characters form any valid string $\longrightarrow b_{n-1}$ choices.
- If it starts with `0`: the next character *must* be `1` (to avoid "00"), and the remaining $n-2$ characters form any valid string $\longrightarrow b_{n-2}$ choices.

This gives the Fibonacci-like recurrence $b_n = b_{n-1} + b_{n-2}$.




**Problem Statement**

How many binary strings of length 10 contain two consecutive zeros?

### Solution

We use **complementary counting**.

**Step 1: Count all binary strings of length 10.**

Each position has 2 choices, so the total is $2^{10} = 1024$.

**Step 2: Count binary strings of length 10 with *no* two consecutive zeros.**

Let $b_n$ denote the number of binary strings of length $n$ with no two consecutive 0s.

**Deriving the recurrence.** Consider a valid string of length $n$:

- If it starts with `1`: the remaining $n-1$ characters can be any valid string of length $n-1$, giving $b_{n-1}$ choices.

- If it starts with `0`: the next character must be `1` (otherwise we have "00"), and the remaining $n-2$ characters can be any valid string of length $n-2$, giving $b_{n-2}$ choices.

Therefore:

$$b_n = b_{n-1} + b_{n-2}, \qquad n \geq 3.$$

**Base cases.**

- $b_1 = 2$: the strings are `0` and `1`.
- $b_2 = 3$: the valid strings are `01`, `10`, `11` (excluding `00`).

**Step 3: Compute $b_n$ for $n = 3, 4, \ldots, 10$.**

| $n$ | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** |
|---|---|---|---|---|---|---|---|---|---|---|
| $b_n$ | 2 | 3 | 5 | 8 | 13 | 21 | 34 | 55 | 89 | 144 |

$b_9 = b_8 + b_7 = 55 + 34 = 89$

Detailed computation:

$$\begin{aligned}
b_3 &= b_2 + b_1 = 3 + 2 &&= 5 \\
b_4 &= b_3 + b_2 = 5 + 3 &&= 8 \\
b_5 &= b_4 + b_3 = 8 + 5 &&= 13 \\
b_6 &= b_5 + b_4 = 13 + 8 &&= 21 \\
b_7 &= b_6 + b_5 = 21 + 13 &&= 34 \\
b_8 &= b_7 + b_6 = 34 + 21 &&= 55 \\
b_9 &= b_8 + b_7 = 55 + 34 &&= 89 \\
b_{10} &= b_9 + b_8 = 89 + 55 &&= 144
\end{aligned}$$

**Observation.** The values $b_n = 2, 3, 5, 8, 13, 21, 34, 55, 89, 144$ are the Fibonacci numbers $F_3, F_4, \ldots, F_{12}$. In general, $b_n = F_{n+2}$ where $F_1 = 1, F_2 = 1, F_3 = 2, \ldots$

**Step 4: Apply complementary counting.**

$$\text{Strings with "00"} = \text{Total} - \text{Strings without "00"} = 1024 - 144 = 880.$$

**Verification for small cases.** For $n = 2$: total $= 4$, strings without "00" $= 3$ (`01`, `10`, `11`), so strings with "00" $= 1$ (namely `00`). ✓

For $n = 3$: total $= 8$, $b_3 = 5$ (`010`, `011`, `101`, `110`, `111`), so strings with "00" $= 3$ (`001`, `100`, `000`). ✓

The number of binary strings of length 10 that contain two consecutive zeros is $\boxed{880}$.